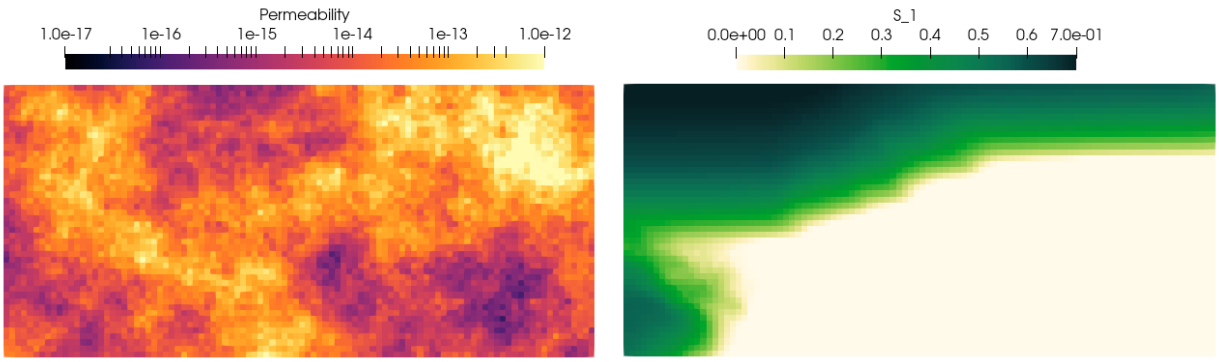
DARSim Example:

2D CCS with Heterogeneity (SPE10 Slice)



This example demonstrates how to use a heterogenous permeability field. In this case, the example uses a 2D slice of permeability field from the SPE10 benchmark problem, as shown above (left). The injection point is placed in the bottom left, and the production well is placed in the top right. After approximately 70 years of injection, the distribution of CO_2 saturation is shown above (right).

To use the heterogenous field, the input file `Sim.txt` defines the permeability file to use (Perm_mSPE10T_v1.txt). In this example, the same file is used for permeability in each direction (x, y, and z), but different files can be used to account for anisotropy.

```
PERMEABILITY_UNIT
      mD

PERMEABILITY_SCALE
      Linear

      INCLUDE
      PERMEABILITY_X
            Perm_mSPE10T_v1.txt

      INCLUDE
      PERMEABILITY_Y
            Perm_mSPE10T_v1.txt

      INCLUDE
      PERMEABILITY_Z
            Perm_mSPE10T_v1.txt
```